

NONCLINICAL REPORT

Nonclinical Report No. MOD-3937

Vaccination and Protein Restimulation in Young BALB/c Mice with Enhanced Respiratory
Disease Endpoint Monitoring

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DATE

05 October 2020

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AA	amino acid
alum	aluminum hydroxide
CoV	coronavirus
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine
ELISA	enzyme-linked immunosorbent assay
ERD	enhanced respiratory disease
GMP	Good Manufacturing Practice
IACUC	Institutional Animal Care and Use Committee
IC ₅₀	half-maximal inhibitory concentration
IFN	interferon
IgG	immunoglobulin G
IL	interleukin
IM	intramuscular
IV	intravenous
LNP	lipid nanoparticle
LOD	limit of detection
OD	optical density
mRNA	messenger RNA
PBS	phosphate-buffered saline
PEG	polyethylene glycol
PEG-2000DMG	1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol
RSV	respiratory syncytial virus
S	spike
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
SARS	severe acute respiratory syndrome
SARS-CoV-2	2019 novel coronavirus
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate
Th1	T-helper 1
Th2	T-helper 2
UTR	untranslated region

QUALITY ASSURANCE STATEMENT

The study was not conducted according to Good Laboratory Practice regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized S protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic.

Despite prior clinical safety of other mRNA vaccines, vaccines against viral respiratory diseases should be evaluated to address the theoretical concern of enhanced respiratory disease (ERD). The potential for ERD is predicted for vaccines where immune responses are characterized by low neutralizing antibody responses and/or a predominant T-helper 2 (Th2) mediated T-cell response and was prompted by preclinical observations in severe acute respiratory syndrome (SARS)-CoV (Bolles et al 2011; Yip et al 2014) and Middle East respiratory syndrome-CoV vaccines (Houser et al 2017); feline infectious peritonitis virus and vaccines in cats (Pedersen et al 2009; Pedersen et al 2012; Takano et al 2008); as well as the enhanced disease seen with respiratory syncytial virus (RSV) (Kim et al 1969), measles, and dengue vaccines in humans (Smatti et al 2018).

The objectives of Study MOD-3937 were to evaluate the immunogenicity of mRNA-1273 Drug Product and to address the theoretical concern of ERD in young, female BALB/c mice relative to SARS-CoV-2 S-2P protein adjuvanted with aluminum hydroxide (alum; (b) (4)).

ModernaTX, Inc. (Sponsor) is developing mRNA-1273, a novel lipid nanoparticle (LNP)-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes for the SARS-CoV-2 S glycoprotein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P). The mRNA-1273 Drug Product consists of the mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to the Sponsor's mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy) hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol (PEG2000-DMG), respectively. The test articles in this study consisted of mRNA-1273 Drug Product (WH2020_NatSP_2P, Tech dev DP), SARS-CoV-2 S-2P protein formulated with alum adjuvant ((b) (4)), and phosphate-buffered saline (PBS) as a control.

Groups of BALB/c mice were vaccinated with 1 µg (n = 7) or 10 µg (n = 8) of mRNA-1273 Drug Product, 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) (n = 8), or PBS (n = 3). Vaccine or control was administered via intramuscular (IM) injection into the hind leg on a prime

(Week 0)/boost (Week 2) schedule. At Week 4 (Day 28), serum was collected from all animals and assessed for immunoglobulin G (IgG) antibody binding to SARS-CoV-2 S protein (Sino Biological) by an enzyme-linked immunosorbent assay (ELISA). Serum samples were also tested for neutralizing activity against homologous SARS-CoV-2 S protein via a pseudotyped lentivirus reporter neutralization assay. Spike protein-specific IgG2a and IgG1 antibody titers were also measured by ELISA, and subclass ratios were calculated. A Th2-directed CD4 T-cell response is characterized in mice by the presence of IgG1 antibody subclass titers and a corresponding lack of IgG2a antibody subclass titers. One day after sera collection (Day 29), the 10 µg SARS-CoV-2 S-2P protein + 250 µg (b) (4) group animals were restimulated with 10 µg of SARS-CoV-2 S-2P protein (without adjuvant), and the PBS group animals were injected with PBS administered via intravenous (IV) injection into the tail vein to activate antigen-specific T-cells. Mice in the 1 µg and 10 µg mRNA-1273 groups were not restimulated. Four days after restimulation (Day 33), spleens were collected and splenocytes isolated from each group of mice were stimulated for 24 hours in vitro to activate antigen-specific splenocytes using 2 S peptide pools that in combination cover the entire SARS-CoV-2 S protein (b) (4) (b) (4). After the 24-hour stimulation with the 2 S peptide pools, culture supernatant was collected to measure the levels of secreted cytokines using (b) (4) analysis.

Binding and neutralizing antibody titers were measured, as high levels of binding antibody with low levels of neutralization activity have been associated with disease enhancement mediated by immune complex deposition and complement activation. Additionally, the balance of T-helper 1 (Th1) and Th2 levels was evaluated. Th2-directed immune responses have been associated with ERD in children following immunization with whole-inactivated virus vaccines against RSV and measles virus (Fulginiti et al 1967; Kim et al 1969). Similar associations were reported in animal models with whole-inactivated and other types of experimental SARS-CoV vaccines (Bolles et al 2011; Czub et al 2005; Corbett et al 2020; Deming et al 2007). Thus, the antibody titers for IgG antibody subclasses (IgG2a and IgG1) were measured, and the ratios of IgG antibody subclasses (IgG2a/IgG1) were calculated; the ratio of IgG antibody subclasses is considered a surrogate measurement to indicate the Th1/Th2 response. A ratio of IgG1 subclass with low or no IgG2a is associated with a Th2-directed immune response, whereas balanced IgG2a/IgG1 antibody subclasses are associated with a Th1-directed immune response. Lastly, levels of secreted cytokines from splenocytes stimulated in vitro with the 2 S peptide pools were evaluated to identify Th1- and Th2-directed immune responses. In mice, Th1-directed cytokine T-cells secrete interferon (IFN)-γ in response to antigen-specific stimulation, whereas Th2 T-cells will produce interleukin (IL)-4, IL-5, and/or IL-13. The ratio between the Th1 and Th2 cytokines indicates the type of T-cell response. A Th1-directed CD4 T-cell profile is expected to have a ratio that is skewed toward IFN-γ.

mRNA-1273 was shown to be immunogenic and capable of driving a Th1-directed response. Vaccination with 1 µg and 10 µg of mRNA-1273 induced similar SARS-CoV-2 S-specific IgG titers and neutralizing activity compared with 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) in serum collected from young, female BALB/c mice at Week 4. A dose-dependent increase in neutralization titers was observed for mRNA-1273, with the 10-µg dose driving a statistically higher response versus the 1-µg dose.

High titers of both IgG1 and IgG2a were detected in mice administered 1 µg and 10 µg of mRNA-1273. In contrast, mice administered 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) had high titers of IgG1 and low titers of IgG2a. A comparison of the IgG2a/IgG1 ratios showed that 1 µg and 10 µg of mRNA-1273 had an antibody subclass profile that correlates to a Th1-directed CD4 T-cell response. This contrasted with the response to 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4), which drove a predominant IgG1 antibody subclass response, a response profile that correlates to a Th2-directed CD4 T-cell response.

Following in vitro stimulation with the 2 S peptide pools, the splenocytes from mRNA-1273-immunized mice secreted more IFN-γ than IL-4, IL-5, or IL-13, indicating a Th1-directed immune response; whereas, 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) induced a Th2-directed response.

Overall, these results demonstrate that, unlike the SARS-CoV-2 S-2P protein adjuvanted with alum, which elicits a Th2-directed response, mRNA-1273 drives a Th1-directed immune response and induces robust binding and neutralization activity, an immune signature not predicted to drive vaccine-associated ERD.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.2. Performing Laboratory

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.3. Study Number

Study number MOD-3937

2.1.4. Purpose of the Study

The objectives of Study MOD-3937 were to evaluate the immunogenicity of mRNA-1273 Drug Product and to address the theoretical concern of ERD in young, female BALB/c mice relative to SARS-CoV-2 S-2P protein adjuvanted with alum.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved and conducted in accordance with the regulations of the Sponsor's Institutional Animal Care and Use Committee (IACUC Protocol 18-03-04).

2.1.6. Study Director

(b) (6)

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.7. Key Personnel

Role	Study MOD-3937
Study coordinators	(b) (6) ModernaTX, Inc. (b) (6) ModernaTX, Inc.
Study execution group lead	(b) (6) ModernaTX, Inc.
Analytical ELISA assays lead	(b) (6) ModernaTX, Inc.
Analytical T-cell assays lead	(b) (6) ModernaTX, Inc.
(b) (4) lead	(b) (6) ModernaTX, Inc.
Report compilation	PPD 3900 Paramount Parkway Morrisville, NC 27560

2.1.8. Dates of Study Performance

Schedule of Events	Study MOD-3937
Test administration Prime dose Boost dose	02 April 2020 (Study Day 1) 16 April 2020 (Study Day 15)
Serum collection	29 April 2020 (Study Day 28)
Restimulation	30 April 2020 (Study Day 29)
Spleen collection	04 May 2020 (Study Day 33)

2.1.9. Records Maintained

Data are archived in the Sponsor Electronic Notebook System at the following location:

Electronic Notebook Project: Moderna Informatics Digital In Vivo Studies Application

Electronic Notebook Experiment: Moderna Informatics Digital In Vivo Studies Application

2.2. Test Articles

The test articles included 1 µg and 10 µg of mRNA-1273 and 10 µg of purified SARS-CoV-2 S-2P protein + 250 µg of (b) (4). Phosphate-buffered saline was used as a negative control.

2.2.1. mRNA-1273 Test Article

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102; cholesterol; DSPC; and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in this study was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP, AMPDP-200005) prepared with the same method as the good manufacturing practice (GMP) mRNA-1273 clinical Drug Product ([Appendix 1](#)). This product is a novel LNP-encapsulated, mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes for the full-length S protein of SARS-CoV-2, modified to introduce 2 proline residues to stabilize the S protein into a prefusion conformation (S-2P). mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process described previously ([Hassett et al 2019](#)).

Preclinical mRNA-1273 vaccine was manufactured by the Sponsor. A sequence-optimized mRNA encoding prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N¹m-pseudouridine ([Nelson et al 2020](#)). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O-methyltransferase (New England Biolabs). The mRNA was purified (b) (4)

(b) (4), buffer exchanged (b) (4), sterile filtered, and kept frozen at -20°C until further use.

mRNA-1273 was encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as previously described ([Hassett et al 2019](#)) (b) (4)

(b) (4)

The mixture was neutralized with tris(hydroxymethyl)aminomethane-hydrochloride, pH 7.5, sucrose was added as a cryoprotectant, and the final solution was sterile filtered. Vials were filled with mRNA-1273 formulated LNP and stored frozen at -70°C until further use. The Drug Product underwent analytical characterization, which included the determination of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin content, and bioburden; the material was deemed acceptable for the in vivo study (Corbett et al 2020).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No.	mRNA Lot No.	Size (nm)	PDI	EE %
mRNA-1273 Drug Product	mRNA SARS-CoV-2 spike protein ^a	(b) (4)				

Abbreviations: EE% = encapsulation efficiency; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; SARS-CoV-2 = 2019 novel coronavirus.

Note: All mRNA test material stocks were formulated in SM-102-containing lipid nanoparticles with a final storage buffer in 10.7 mM sodium acetate; 8.7% sucrose; 20 mM tris(hydroxymethyl)aminomethane hydrochloride at pH 7.5 and stored frozen at -70°C until further use.

The EE% is the percentage of mRNA (b) (4)

(b) (4).

^a Full-length SARS-CoV-2 spike protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) and including a transmembrane domain for insertion into the cell membrane after translation.

^b Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. SARS-CoV-2 S-2P Protein

Purified SARS-CoV-2 S-2P protein was prepared by the Sponsor. EXPI293 cells (b) (4) were transfected with WH2020_NatSP_2P_RRAR-GSGG-Fd-tHS plasmid DNA (Gene Immune), and then S-2P protein purification and quantification was performed.

2.2.3. (b) (4)

(b) (4) (aluminum hydroxide (b) (4)) (4) was purchased commercially.

2.2.4. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4) was purchased commercially.

2.3. Test System

2.3.1. Mice

Female BALB/c mice, approximately 8 to 9 weeks old, were obtained from Charles River Laboratories (Kingston, New York).

2.4. Methods

2.4.1. Study Design and Treatment Groups

Groups of BALB/c mice were vaccinated with 1 µg or 10 µg of mRNA-1273 Drug Product, 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) or PBS. Vaccine or PBS control was administered via IM injection into the hind leg on a prime (Week 0)/boost (Week 2) schedule (Table 2). At Week 4 (Day 28), serum was collected from all animals and assessed for antibody binding to SARS-CoV-2 S protein (Sino Biological) by an ELISA. Serum samples were also tested for neutralizing activity against homologous SARS-CoV-2 S protein via a pseudotyped lentivirus reporter neutralization assay. Spike protein-specific IgG2a and IgG1 antibody titers were also measured by ELISA, and IgG2a to IgG1 subclass ratios were calculated. One day after sera collection (Day 29), the 10 µg SARS-CoV-2 S-2P protein + 250 µg (b) (4) group animals were restimulated with 10 µg of SARS-CoV-2 S-2P protein (without adjuvant), and the PBS group animals were injected with PBS administered via IV injection into the tail vein to activate antigen-specific T-cells. Mice in the 1 µg and 10 µg mRNA-1273 groups were not restimulated. Four days after restimulation (Day 33), spleens were collected and splenocytes isolated from each group of mice were stimulated for 24 hours in vitro using 2 S peptide pools (S1 and S2) to activate antigen-specific splenocytes. The S1 and S2 peptide pools, in combination, represent peptides with overlapping amino acid (AA) sequences (15mers with 11 AA overlap) that cover the entire SARS-CoV-2 S protein. Culture supernatant was collected after 24 hours and (b) (4) analysis was performed to measure the levels of secreted cytokines in order to evaluate the Th1/Th2 responses.

Table 2: Treatment Regimen for Study MOD-3937

Young Female BALB/c Mice Treatment Groups	No. of Animals	Vaccination Schedule	Sample Collection	Restimulation Schedule	Endpoint Monitoring
10 µg mRNA-1273	8	<u>Prime</u> Week 0/Day 1 <u>Boost</u> Week 2/Day 15	Week 4/Day 28 (sera for ELISA titer measurements and neutralization assay) Week 4/Day 33 (splens for cytokine panel)	N/A	ELISA titer measurements (including IgG, IgG1, and IgG2a); pseudovirus neutralization assay; cytokine panel on overnight culture supernatant
1 µg mRNA-1273	7				
10 µg SARS-CoV-2 S-2P + 250 µg (b) (4)	8			Week 4/Day 29 ^a	
PBS	3			Week 4/Day 29 ^b	

Abbreviations: ELISA = enzyme-linked immunosorbent assay; IgG = immunoglobulin G; N/A = not applicable; PBS = phosphate-buffered saline; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus.

^a Mice were restimulated with 10 µg SARS-CoV-2 S-2P protein (without a djuvant) administered intravenously on Day 29.

^b Mice injected intravenously with PBS on Day 29.

2.4.2. Enzyme-Linked Immunosorbent Assay (ELISA)

The 96-well microtiter plates (b) (4) were coated with 100 µL per well of 1 µg/mL SARS-CoV-2 S protein (S1 + S2, Sino Biological, #40589-V08B1). After overnight incubation at 4°C, plates were washed 4 times using PBS with 0.05% Tween-20 and blocked for 2 hours at 37°C (b) (4). After washing, 5-fold serial dilutions of mouse serum were added (assay diluent-PBS + 0.05% Tween-20 + 5% goat serum). Plates were incubated for 2 hours at 37°C, washed, and then horseradish peroxidase-conjugated goat anti-mouse IgG (b) (4) goat anti-mouse IgG1 (b) (4) or goat anti-mouse IgG2a (b) (4) was added at a 1:10,000 dilution in assay diluent. Plates were incubated for 1 hour at 37°C, washed, and then tetramethylbenzidine substrate (b) (4) (b) (4) was used to detect bound antibody. After incubation at room temperature for 10 minutes for IgG and IgG1 and 20 minutes for IgG2a, the reaction was stopped by adding tetramethylbenzidine stop solution (b) (4) and the absorbance was measured at optical density (OD) 450 nm using a (b) (4) plate reader.

2.4.3. Lentivirus-Based Pseudovirus Neutralization Assay

(b) (4)

(b) (4)

(b) (4)

and then pseudovirus neutralization assays were performed (Pallesen et al 2017). Briefly, heat-inactivated serum was mixed with pseudoviruses, incubated, and then added to angiotensin-converting enzyme 2 receptor-expressing HEK293T cells obtained from (b) (6) at Scripps Research Institute. Cells were lysed after 72 hours of incubation, and luciferase activity (relative light units) was measured using a (b) (4) Luminometer by (b) (4) (Corbett et al 2020).

2.4.4. T-cell Stimulation and Cytokine Analysis

A gentle (b) (4) tissue dissociator and tissue dissociation kit (b) (4)) was used to generate single cell splenocyte suspensions from whole mouse spleens. Following tissue dissociation, the cells were sieved through a 70 μm filter, and then density gradient centrifugation was performed using Fico/Lite-LM medium (b) (4)) to enrich for peripheral blood mononuclear cells. Then, 2×10^6 splenocytes/well (96-well plate) were stimulated in vitro with 2 S peptide pools (S1 and S2), which in combination represent peptides with overlapping AA sequences (15mers with 11 AA overlap) that cover the entire SARS-CoV-2 S protein; the peptides were split into 2 pools to increase stability. Both peptide pools were used at a final concentration of 1 $\mu\text{g/mL}$. For cytokine detection, the cells were incubated with S1 and S2 peptide pools for 24 hours at 37°C, then the plates were centrifuged, and supernatant was collected and frozen at -80°C. After a 2-fold dilution of supernatants, a multiplex bead-based technology (b) (4)) assay was performed using a murine 35-plex kit (b) (4)) with a (b) (4) instrument (b) (4)) to measure the concentration of secreted cytokines (Corbett et al 2020).

2.5. Data Presentation and Analysis

Geometric means or means are represented by the heights of bars or symbols, and error bars represent the corresponding SD. The limit of detection (LOD) for each assay was the starting serum dilution; for example, if serum was diluted 1:25, then the LOD was 25. Dotted lines indicate assay LODs. Mann-Whitney tests were used to compare 2 experimental groups, and Wilcoxon signed-rank tests were used to compare the same animals at different time points. To compare more than 2 experimental groups, Kruskal-Wallis analysis of variance with Dunn's multiple comparisons tests were applied. For antibody responses, titers were determined using a 4-parameter logistic curve fit in GraphPad Prism (GraphPad Software, Inc.) and defined as the reciprocal dilution at approximately $\text{OD}_{450\text{nm}} = 1.0$ (normalized to a mouse standard on each plate). A Spearman's correlation was used to correlate binding antibody titers to neutralizing antibody titers (* = p value < .05). Percent neutralization was normalized considering uninfected cells as 100% neutralization and cells infected with only pseudovirus as 0% neutralization. Half maximal

inhibitory concentration (IC_{50}) titers were determined using a log (agonist) versus normalized response (variable slope) nonlinear function in Prism v8 (GraphPad) ([Corbett et al 2020](#)).

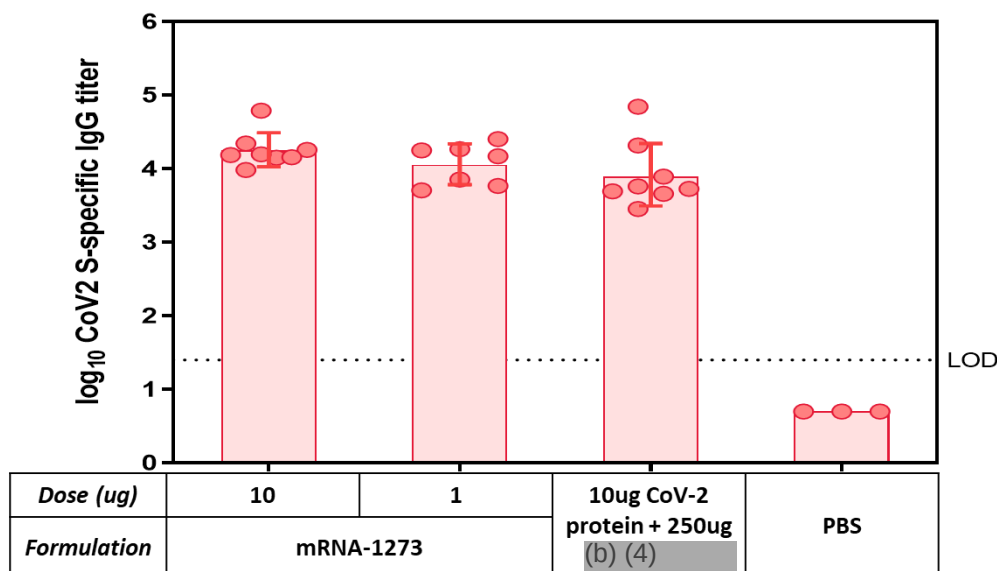
3. RESULTS

3.1. IgG Titers and Neutralizing Activity in Sera From mRNA-1273-immunized Mice

Two weeks post-boost (Day 29), serum was analyzed for IgG titers, neutralizing activity, and IgG subclass (IgG2a and IgG1) titers from young, female BALB/c mice vaccinated with 1 µg and 10 µg of mRNA-1273 and 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4).

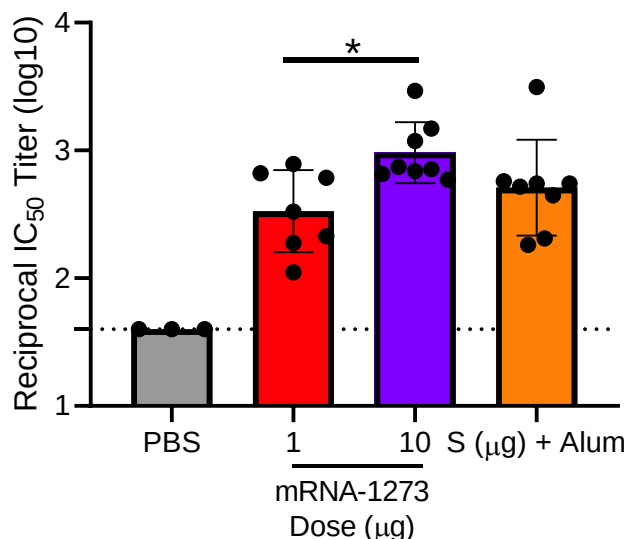
Vaccination with mRNA-1273 at 1 µg and 10 µg induced similar SARS-CoV-2 S-specific IgG titers (Figure 1) and neutralizing activity (Figure 2) compared with 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4). A dose-dependent increase in neutralization titers was observed for mRNA-1273, with the 10-µg dose driving a statistically higher response versus the 1-µg dose.

Figure 1: Serum IgG Titers at Day 28



Abbreviations: CoV-2 = SARS-CoV-2; CoV-2 protein = SARS-CoV-2 S-2P protein; IgG = immunoglobulin G; LOD = limit of detection; PBS = phosphate-buffered saline; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus.

Dotted line indicates the assay limit of detection.

Figure 2: Serum Neutralizing Antibodies Against SARS-CoV-2 Pseudoviruses at Day 28

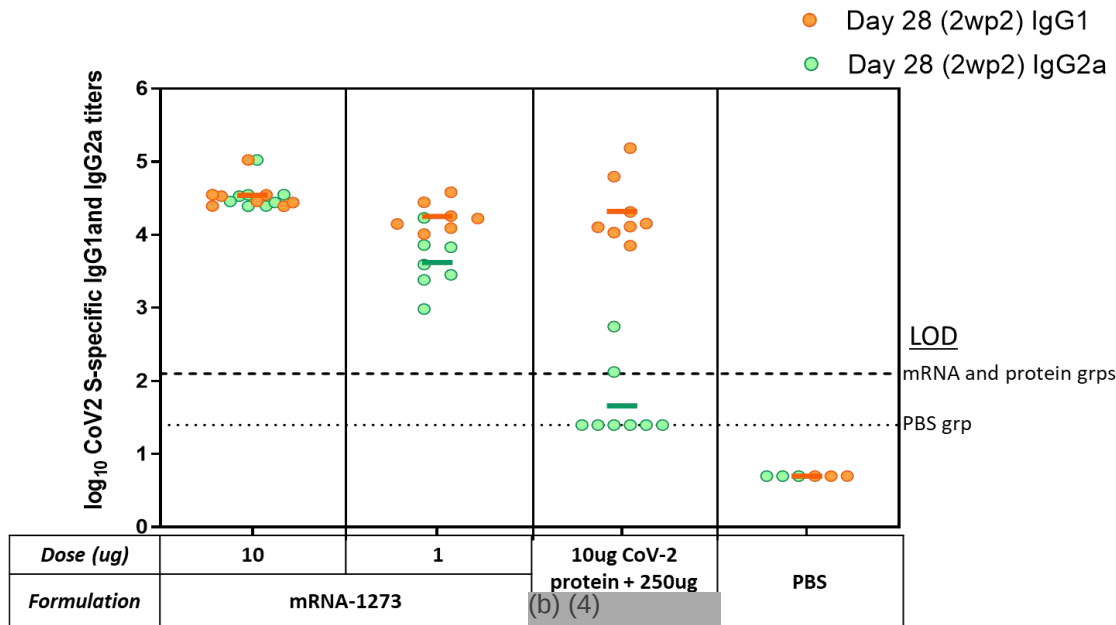
Abbreviations: Alum = aluminum hydroxide; IC₅₀ = half maximal inhibitory concentration; PBS = phosphate-buffered saline; S = SARS-CoV-2 S-2P protein; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus.

Dotted line indicates the assay limit of detection.

* = *p* value < .05.

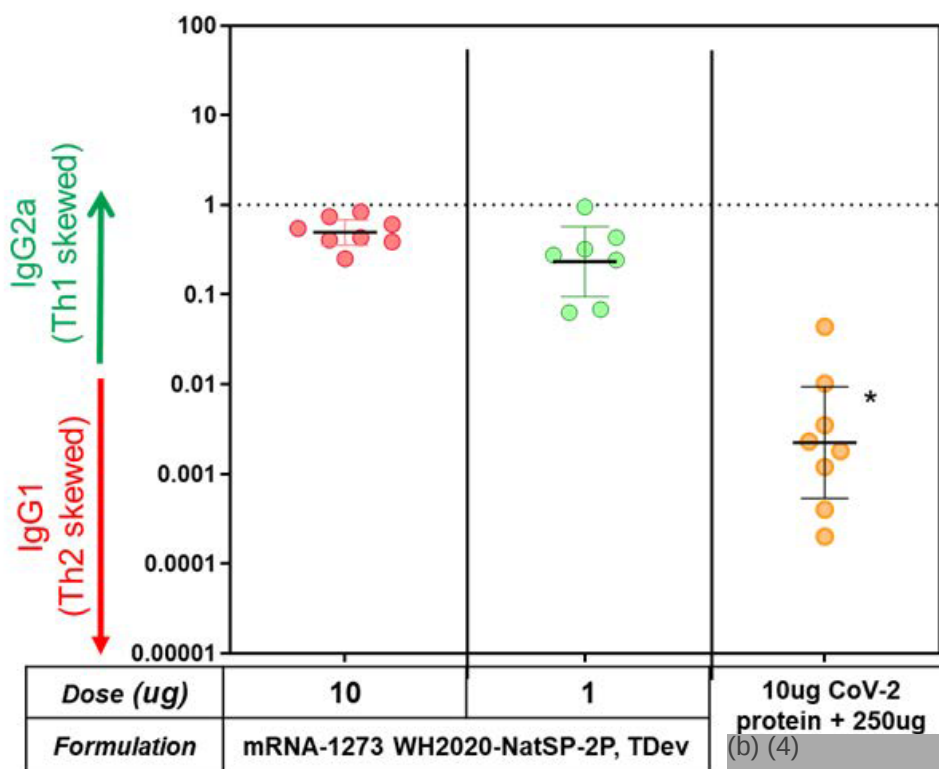
In mice, the ratio of IgG antibody subclasses can be considered a surrogate measurement to indicate the Th1/Th2 response. A ratio of IgG1 subclass with low or no IgG2a is associated with a Th2-directed response, whereas balanced IgG2a/IgG1 antibody subclasses are associated with a Th1-directed response.

High titers of both IgG1 and IgG2a were detected in mice administered 1 μg and 10 μg of mRNA-1273 (Figure 3). In contrast, mice administered 10 μg of SARS-CoV-2 S-2P protein + 250 μg of (b) (4) had high titers of IgG1 and low titers of IgG2a. A comparison of the IgG2a/IgG1 ratios showed 1 μg and 10 μg of mRNA-1273 had a Th1-directed response, whereas 10 μg of SARS-CoV-2 S-2P protein + 250 μg of (b) (4) had a Th2-directed response (predominantly IgG1) (Figure 4).

Figure 3: Serum IgG1 and IgG2a Titers at Day 28

Abbreviations: 2wp2 = 2 weeks post-boost; CoV2 = SARS-CoV-2; CoV-2 protein = SARS-CoV-2 S-2P protein; grps = groups; IgG = immunoglobulin G; LOD = limit of detection; PBS = phosphate-buffered saline; S = spike; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus.

Dotted line indicates the assay limit of detection.

Figure 4: Serum IgG2a/IgG1 Titer Ratios at Day 28

Abbreviations: CoV-2 protein = SARS-CoV-2 S-2P protein; IgG = immunoglobulin G; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus.

Dotted line indicates the assay limit of detection.

*Note: Six of 8 samples had IgG2a titers < 125 (below limit of detection); therefore, the IgG2a:IgG1 ratio is likely much lower than indicated on the graph.

3.2. Cytokine Activity Levels in Supernatant of In Vitro Splenocytes From mRNA-1273-immunized Mice Following Stimulation With Spike Protein Peptide Pools

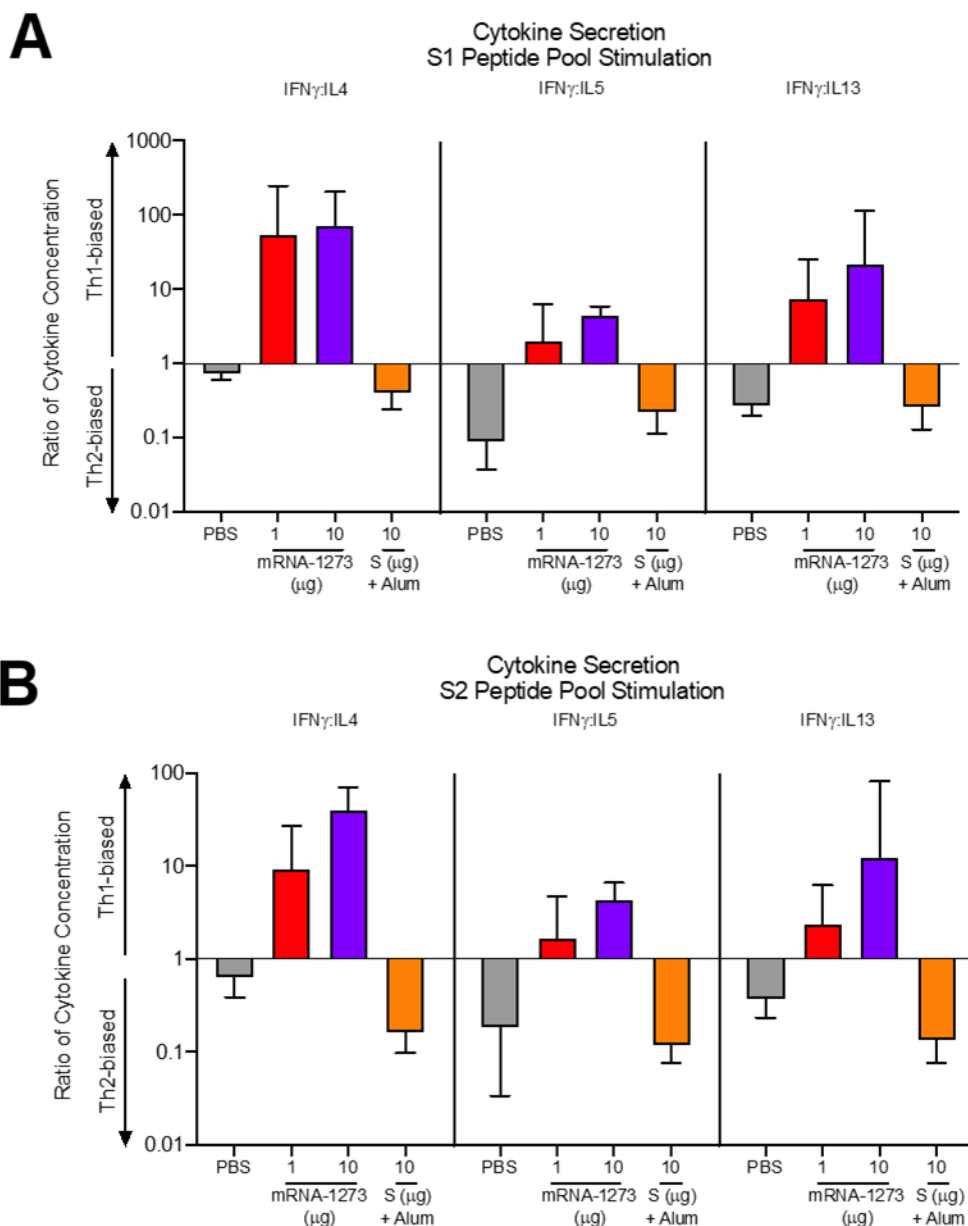
One day after serum collection (Day 29), the 10 µg SARS-CoV-2 S-2P protein + 250 µg (b) (4) group animals were restimulated with 10 µg of SARS-CoV-2 S-2P protein (without adjuvant), and the PBS group animals were injected with PBS to activate antigen-specific T-cells. Four days after restimulation (Day 33), spleens were collected and splenocytes isolated from each group of mice were stimulated for 24 hours in vitro with 2 S peptide pools. Culture supernatant was collected after 24 hours to measure the levels of secreted cytokines in order to evaluate the Th1/Th2 responses.

In mice, Th1-directed cytokine T-cells secrete IFN-γ in response to antigen-specific stimulation, whereas Th2 T-cells will produce IL-4, IL-5, and/or IL-13. The ratio between the Th1 and Th2

cytokines indicates the type of T-cell response. A Th1-directed CD4 T-cell profile is expected to have a ratio that is skewed toward IFN- γ .

Following in vitro stimulation with the 2 S peptide pools corresponding to the SARS-CoV-2 S protein, splenocytes from mRNA-1273-immunized mice secreted more IFN- γ than IL-4, IL-5, or IL-13, indicating a Th1-directed immune response; whereas, 10 μ g of SARS-CoV-2 S-2P protein + 250 μ g of (b) (4) induced Th2-directed cytokine secretion ([Figure 5](#)).

Figure 5: Cytokine Concentrations in Supernatant of In Vitro Splenocytes on Day 33 Following Stimulation With Spike Protein Peptide Pools



Abbreviations: Alum = aluminum hydroxide; IFN = interferon; IL = interleukin; PBS = phosphate-buffered saline; S = SARS-CoV-2 S-2P protein; S-2P = spike protein with 2 proline substitutions within the heptad repeat 1 domain; SARS-CoV-2 = 2019 novel coronavirus; Th1 = T-helper 1; Th2 = T-helper 2.

4. CONCLUSIONS

mRNA-1273 was shown to be immunogenic and capable of driving a Th1-directed response. Vaccination with 1 µg and 10 µg of mRNA-1273 induced similar SARS-CoV-2 S-specific IgG titers and neutralizing activity compared with 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) in serum collected from young, female BALB/c mice at Week 4. A dose-dependent increase in neutralization titers was observed for mRNA-1273, with the 10-µg dose driving a statistically higher response versus the 1-µg dose.

High titers of both IgG1 and IgG2a were detected in mice administered 1 µg and 10 µg of mRNA-1273. In contrast, mice administered 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) had high titers of IgG1 and low titers of IgG2a. A comparison of the IgG2a/IgG1 ratios showed that 1 µg and 10 µg of mRNA-1273 had an antibody subclass profile that correlates to a Th1-directed CD4 T-cell response. This contrasted with the response to 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4), which drove a predominant IgG1 antibody subclass response, a response profile that correlates to a Th2-directed CD4 T-cell response.

Following in vitro stimulation with the 2 S peptide pools, the splenocytes from mRNA-1273-immunized mice secreted more IFN-γ than IL-4, IL-5, or IL-13, indicating a Th1-directed immune response; whereas, 10 µg of SARS-CoV-2 S-2P protein + 250 µg of (b) (4) induced a Th2-directed response.

Overall, these results demonstrate that, unlike the SARS-CoV-2 S-2P protein adjuvanted with alum, which elicits a Th2-directed response, mRNA-1273 drives a Th1-directed immune response and induces robust binding and neutralization activity, an immune signature not predicted to drive vaccine-associated ERD.

5. REFERENCES

- Bolles M, Deming D, Long K, Agnihothram S, Whitmore A, Ferris M, et al. A double-inactivated severe acute respiratory syndrome coronavirus vaccine provides incomplete protection in mice and induces increased eosinophilic proinflammatory pulmonary response upon challenge. *J Virol*. 2011;85(23):12201-15.
- Böttcher E, Matrosovich T, Beyerle M, Klenk HD, Garten W, Matrosovich M. Proteolytic activation of influenza viruses by serine proteases TMPRSS2 and HAT from human airway epithelium. *J Virol*. 2006;80(19):9896-8.
- Corbett KS, Edwards D, Leist SR, Abiona OM, Boyoglu-Barnum S, Gillespie RA et al. SARS-CoV-2 mRNA vaccine design enabled by prototype pathogen preparedness. *Nature*. 2020 Aug 5;10.1038/s41586-020-2622-0. Epub 2020 Aug 5.
- Czub M, Weingartl H, Czub S, He R, Cao J. Evaluation of modified vaccinia virus Ankara based recombinant SARS vaccine in ferrets. *Vaccine*. 2005;23(17-18):2273-9.
- Deming D, Sheahan T, Heise M, Yount B, Davis N, Sims A, et al. Vaccine efficacy in senescent mice challenged with recombinant SARS-CoV bearing epidemic and zoonotic spike variants. *PLoS Med*. 2006;3(12):e525. Erratum in: *PLoS Med*. 2007;4(2):e80.
- Fulginiti VA, Eller JJ, Downie AW, Kempe CH. Altered reactivity to measles virus. Atypical measles in children previously immunized with inactivated measles virus vaccines. *JAMA*. 1967;202(12):1075-80.
- Hassett KJ, Benenato KE, Jacquinet E, Lee A, Woods A, Yuzhakov O, et al. Optimization of Lipid Nanoparticles for Intramuscular Administration of mRNA Vaccines. *Mol Ther Nucleic Acids*. 2019;15:1-11.
- Houser KV, Broadbent AJ, Gretebeck L, Vogel L, Lamirande EW, Sutton T, et al. Enhanced inflammation in New Zealand white rabbits when MERS-CoV reinfection occurs in the absence of neutralizing antibody. *PLoS Pathog*. 2017;13(8):e1006565.
- Kim HW, Canchola JG, Brandt CD, Pyles G, Chanock RM, Jensen K, et al. Respiratory syncytial virus disease in infants despite prior administration of antigenic inactivated vaccine. *Am J Epidemiol*. 1969;89(4):422-34.
- Nelson J, Sorensen EW, Mintri S, Rabideau AE, Zheng W, Besin G, et al. Impact of mRNA chemistry and manufacturing process on innate immune activation. *Sci Adv*. 2020 Jun 24;6(26):eaaz6893.

Pallesen J, Wang N, Corbett KS, Wrapp D, Kirchdoerfer RN, Turner HL, et al. Immunogenicity and structures of a rationally designed prefusion MERS-CoV spike antigen. *Proc Natl Acad Sci U S A*. 2017;114(35):E7348-E7357.

Pedersen NC, Liu H, Dodd KA, Pesavento PA. Significance of coronavirus mutants in feces and diseased tissues of cats suffering from feline infectious peritonitis. *Viruses*. 2009;1(2):166-84.

Pedersen NC, Liu H, Scarlett J, Leutenegger CM, Golovko L, Kennedy H, et al. Feline infectious peritonitis: role of the feline coronavirus 3c gene in intestinal tropism and pathogenicity based upon isolates from resident and adopted shelter cats. *Virus Res*. 2012;165(1):17-28.

Smatti MK, Al Thani AA, Yassine HM. Viral-Induced Enhanced Disease Illness. *Front Microbiol*. 2018;9:2991.

Takano T, Kawakami C, Yamada S, Satoh R, Hohdatsu T. Antibody-dependent enhancement occurs upon re-infection with the identical serotype virus in feline infectious peritonitis virus infection. *J Vet Med Sci*. 2008;70(12):1315-21.

Wang L, Shi W, Joyce MG, Modjarrad K, Zhang Y, Leung K, et al. Evaluation of candidate vaccine approaches for MERS-CoV. *Nat Commun*. 2015;6:7712.

Yip MS, Leung NHL, Cheung CY, Li PH, Lee HHY, Daëron M, et al. Antibody-dependent infection of human macrophages by severe acute respiratory syndrome coronavirus. *Virol J*. 2014;11:82.

6. REPORT APPROVAL

Study Director:

See esignature and date signed on last
page of document

(b) (6)

ModernaTX, Inc.

Date

7. **APPENDICES**

[Appendix 1 Summary of Analysis: Lot AMPDP-200005 Release](#)

APPENDIX 1: SUMMARY OF ANALYSIS: LOT AMPDP-200005 RELEASE



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SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	(b) (4)
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	(b) (4)
RNA Purity	DPTM-0046	(b) (4)	(b) (4)
% Encapsulation	DPTM-0063	(b) (4)	(b) (4)
Filtered Mean Particle Size	DPTM-0012	(b) (4)	(b) (4)
Filtered Polydispersity Index		(b) (4)	(b) (4)



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Test	Method	Target Attributes	Results
Lipid Identification:			
SM-102	DPTM-0019	(b) (4)	
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Content (mg/mL)			
SM-102	DPTM-0019	(b) (4)	(b) (4)
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Impurities (% Area)			
Individual Impurities	DPTM-0019	(b) (4)	
Total Impurities			
pH	USP <791> DPTM-0028		
Osmolality	USP <785> DPTM-0029		
Bacterial Endotoxin	USP <85>		
Bioburden	USP<61>		
Particulate Matter	USP <788> Method 2		



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REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval

Signature Page for VV-NC-000832 v1.0

Approval	(b) (6) 11-Oct-2020 22:48:44 GMT+0000
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